Michael Fischbach

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Top Secret Clearance (In Process) | SCI Eligible | Malibu, CA

EXPERIENCE

WERC Design Contest Team Lead – NASA Artemis Wireless Capacitive Charger 2026 – Present
- Lead a 12-person graduate/undergraduate team developing a 1 kW capacitive charger for NASA's Artemis lunar rover
- Set schedule and integrate power, coupler, controls, and structures; develop plate-based alignment sensing for rover docking
- Targets: 80%+ efficiency across a 100 mm gap at full scale, 25 W at 70%+ on the bench; dust, vacuum, -203°C to 54°C

All Domain Integration – Mechanical Engineering Intern 2025 – Present
- **SCI eligible · Submitted for Top Secret clearance**
- Support mission systems engineers on Space Force and Air Force efforts as part of a SETA contractor team
- Use locally hosted large language models (LLMs) to analyze satellite sensor data and draw knowledge-based conclusions

Chameleon Ramps – Founder & CEO 2019 – 2026
- Designed 40+ concrete fingerboard obstacles ($10–$50 each); proprietary molding process cut material waste ~40%
- Generated $39,600 in revenue across 627 orders by running a design-test-refine loop of 4–6 prototype generations per product, modeling in Fusion 360 and 3D printing prototypes to build robust, reusable molds for repeat production
- Grew revenue 30% via a 7-creator sponsorship program; co-designed Recycled Ramps collaboration; shipped to 15 countries

EDUCATION

California Polytechnic State University, San Luis Obispo Sep 2025 – Expected May 2028
B.S. Mechanical Engineering, Honors Program | GPA 3.79 | Cal Poly Golf Engineers & Agentic Engineering
Coursework: Subtractive Manufacturing, Materials Joining, Materials Engineering; in progress: Statics, Solid Modeling
- Designed a low-cost, accessible door handle for one-handed users in Cal Poly's Red Brick dorms; PLA base, TPU head
- Built multi-component mechanisms (piston, Geneva wheel, CNC-machined screwdriver) with tolerance stack-up and GD&T

ENGINEERING PROJECTS

CNC-Milled Golf Putter – Full CAD-to-Part Build 2025
3-axis Haas CNC, Fusion 360, DFM
- Designed and machined a novel putter from 6061 in two ops with adaptive toolpaths validated with 12 PLA prototypes
- Modeled unit cost at $40.18 vs. $250 retail; 1 of 8 finalists (40+ students), presented to 200+ alumni at Capstone Night

Apricot Lane Farms – Net-Zero Carbon Shipping Container Space 2024
- Selected as 1 of 3 students to design a net-zero carbon workspace for Apricot Lane Farms; farm chose the L-shaped design
- Analyzed structural integrity of container walls with cutouts and roof overhang; assessed solar power needs with the farmers
- Drafted the floor plan in AutoCAD with an ADA-accessible restroom and incinerating toilet; built a 1:10-scale model

Wave Energy Converter – Team Lead 2023
Bridgeport mill, Fusion 360, fluid dynamics
- Led a five-person team designing and machining a 1:12-scale rack-and-pinion buoy-anchor converter, all parts made in-house; prototype produced 15 V per motor, projecting 360 V at full scale

TECHNICAL SKILLS

CAD / CAE: Fusion 360 (6+ years, 55+ designs), SolidWorks (2+ years), AutoCAD, Autodesk CFD
Manufacturing: 3-axis Haas CNC machining, Bridgeport mill, lathe, FDM/SLA 3D printing, composite & concrete casting
Methods: GD&T (ASME Y14.5), DFM, tolerance stack-up analysis, iterative prototyping, cost modeling
AI: Claude Code, Codex, and Gemini via agentic workflows in Antigravity IDE | **Software:** Python, Git, MATLAB, Linux
Certification: Haas CNC Certification Program – Basic Mill Operator, online coursework completed (Haas Automation, 2026)

INTERESTS

Backpacking – 500 mi | **Senior Patrol Leader & Eagle Scout** – BSA, 2025 | **Weightlifting** – 355 lb squat
Channel Islands Rowing Club – 8 podium placements | **Poker** – Cal Poly Poker Tournament 3rd of 126
Watercolor/Drawing – 3 contest wins | **Volunteering** – Meal Outreach (5+ years, St. Aidan's Church)